

Analytical Bias and Variance Correction for Quantized Linear and Convolutional Layers

ACIQ project (derivation, after Nagel et al. 2019)

1 Setup and Notation

We consider a single weight-bearing layer that produces pre-activation output $\mathbf{y}$ from input $\mathbf{x}$ using FP32 weights $\mathbf{W}$ and bias $\mathbf{b}$. After post-training quantization the same layer uses quantized weights $\widetilde{\mathbf{W}}$. Bias is kept in FP32. We define the per-element *quantization error*

$$\boldsymbol{\epsilon} = \widetilde{\mathbf{W}} - \mathbf{W}, \quad (1)$$

so that

$$\widetilde{\mathbf{y}} = \widetilde{\mathbf{W}}\mathbf{x} + \mathbf{b} = \mathbf{W}\mathbf{x} + \boldsymbol{\epsilon}\mathbf{x} + \mathbf{b} = \mathbf{y} + \boldsymbol{\epsilon}\mathbf{x}. \quad (2)$$

We assume Batch Normalization (BN) is folded into the preceding weight layer, but that the BN scale γ and shift β have been captured before fusion. The activation function is ReLU. Under the assumption made in [1] that the pre-ReLU output of a BN-folded layer is approximately Gaussian, the per-channel input distribution to a downstream layer follows a *ReLU-clipped normal* with parameters (β_c, γ_c^2) .

Throughout this document we use:

- c_o, c_i for output / input channel indices.
- m, n for kernel spatial indices.
- $\mathbb{E}[\mathbf{x}_{c_i}]$ and $\text{Var}[\mathbf{x}_{c_i}]$ for the per-channel mean and variance of the layer input, assumed spatially stationary (i.i.d. across spatial positions within a channel).
- Independence *between* input channels (a standard analytical simplification; the same as [1]).

2 Bias Correction — Linear Layer

For a fully-connected layer [1, Eq. 16–17],

$$\mathbb{E}[\mathbf{y}] = \mathbb{E}[\mathbf{y}] + \mathbb{E}[\boldsymbol{\epsilon}\mathbf{x}] - \mathbb{E}[\boldsymbol{\epsilon}\mathbf{x}] \quad (3)$$

$$= \mathbb{E}[\widetilde{\mathbf{y}}] - \mathbb{E}[\boldsymbol{\epsilon}\mathbf{x}], \quad (4)$$

so subtracting $\mathbb{E}[\boldsymbol{\epsilon}\mathbf{x}] = \boldsymbol{\epsilon}\mathbb{E}[\mathbf{x}]$ from $\widetilde{\mathbf{y}}$ recovers the FP32 mean. Per output unit c_o ,

$$\Delta b_{c_o} = \sum_{c_i} \epsilon_{c_o, c_i} \mathbb{E}[\mathbf{x}_{c_i}]. \quad (5)$$

The correction is folded into the bias:

$$\mathbf{b}_{c_o}^{\text{corr}} = \mathbf{b}_{c_o} - \Delta b_{c_o}. \quad (6)$$

3 Bias Correction — Convolutional Layer with ReLU

For a 2D convolution $\mathbf{y}_{c_o,i,j} = \sum_{c_i,m,n} \mathbf{W}_{c_o,c_i,m,n} \mathbf{x}_{c_i,i-m,j-n} + \mathbf{b}_{c_o}$, the same decomposition gives $\mathbb{E}[\epsilon * \mathbf{x}]$ pointwise. Under spatial stationarity $\mathbb{E}[\mathbf{x}_{c_i,i-m,j-n}] = \mathbb{E}[\mathbf{x}_{c_i}]$:

$$[\epsilon * \mathbb{E}[\mathbf{x}]]_{c_o,i,j} = \sum_{c_i,m,n} \mathbb{E}[\mathbf{x}_{c_i,i-m,j-n}] \epsilon_{c_o,c_i,m,n} \quad (7)$$

$$= \sum_{c_i} \left[\mathbb{E}[\mathbf{x}_{c_i}] \sum_{m,n} \epsilon_{c_o,c_i,m,n} \right]. \quad (8)$$

This is independent of the spatial location (i, j) and can be folded into the per-output-channel bias. In vector form, with $\epsilon_\Sigma \in \mathbb{R}^{C_o \times C_i}$ defined by $[\epsilon_\Sigma]_{c_o,c_i} = \sum_{m,n} \epsilon_{c_o,c_i,m,n}$,

$$\boxed{\Delta \mathbf{b} = \epsilon_\Sigma \mathbb{E}[\mathbf{x}], \quad \mathbf{b}^{\text{corr}} = \mathbf{b} - \Delta \mathbf{b}.} \quad (9)$$

Since (6) is the special case $K = 1$ of (9) (or equivalently $\epsilon_\Sigma = \epsilon$ for a fully-connected layer), the implementation handles both with a single contraction.

4 Variance Correction — Linear Layer

The paper of Nagel et al. corrects only the mean. However, weight quantization perturbs the per-channel variance as well. Under per-channel input independence,

$$\text{Var}[\mathbf{y}_{c_o}] = \sum_{c_i} \mathbf{W}_{c_o,c_i}^2 \text{Var}[\mathbf{x}_{c_i}], \quad \text{Var}[\tilde{\mathbf{y}}_{c_o}] = \sum_{c_i} \tilde{\mathbf{W}}_{c_o,c_i}^2 \text{Var}[\mathbf{x}_{c_i}]. \quad (10)$$

Define the per-output-channel rescaling

$$s_{c_o} = \sqrt{\text{Var}[\mathbf{y}_{c_o}] / \text{Var}[\tilde{\mathbf{y}}_{c_o}]} \quad (11)$$

We apply a *mean-preserving* affine rescaling so that the second moment matches FP32 without disturbing whatever mean is in place:

$$\tilde{\mathbf{y}}_{c_o}^{\text{corr}} = s_{c_o} (\tilde{\mathbf{y}}_{c_o} - \mathbb{E}[\tilde{\mathbf{y}}_{c_o}]) + \mathbb{E}[\tilde{\mathbf{y}}_{c_o}] = s_{c_o} \tilde{\mathbf{y}}_{c_o} + (1 - s_{c_o}) \mathbb{E}[\tilde{\mathbf{y}}_{c_o}]. \quad (12)$$

Folding s_{c_o} into the per-channel weight row and into the bias gives:

$$\boxed{\tilde{\mathbf{W}}_{c_o}^{\text{corr}} = s_{c_o} \tilde{\mathbf{W}}_{c_o}, \quad \mathbf{b}_{c_o}^{\text{corr}} = s_{c_o} \mathbf{b}_{c_o} + (1 - s_{c_o}) \mathbb{E}[\tilde{\mathbf{y}}_{c_o}].} \quad (13)$$

Here $\mathbb{E}[\tilde{\mathbf{y}}_{c_o}] = \sum_{c_i} \tilde{\mathbf{W}}_{c_o,c_i} \mathbb{E}[\mathbf{x}_{c_i}] + \mathbf{b}_{c_o}$ is the (uncorrected) quantized mean.

5 Variance Correction — Convolutional Layer with ReLU

Let $\tilde{\mathbf{W}}_\Sigma^{(2)} \in \mathbb{R}^{C_o \times C_i}$ with $[\tilde{\mathbf{W}}_\Sigma^{(2)}]_{c_o,c_i} = \sum_{m,n} \tilde{\mathbf{W}}_{c_o,c_i,m,n}^2$, and similarly for $\mathbf{W}_\Sigma^{(2)}$. Under the same independence and spatial-i.i.d. assumptions used by [1, App. B],

$$\text{Var}[\mathbf{y}_{c_o,i,j}] = \sum_{c_i} [\mathbf{W}_\Sigma^{(2)}]_{c_o,c_i} \text{Var}[\mathbf{x}_{c_i}], \quad (14)$$

which is independent of the spatial location (i, j) . Defining s_{c_o} as before, the same folding as (13) holds, with the convention that s_{c_o} broadcasts across the kernel spatial axes:

$$\boxed{\tilde{\mathbf{W}}_{c_o,c_i,m,n}^{\text{corr}} = s_{c_o} \tilde{\mathbf{W}}_{c_o,c_i,m,n}, \quad \mathbf{b}_{c_o}^{\text{corr}} = s_{c_o} \mathbf{b}_{c_o} + (1 - s_{c_o}) \mathbb{E}[\tilde{\mathbf{y}}_{c_o}],} \quad (15)$$

with $\mathbb{E}[\tilde{\mathbf{y}}_{c_o}] = \sum_{c_i} [\tilde{\mathbf{W}}_\Sigma]_{c_o,c_i} \mathbb{E}[\mathbf{x}_{c_i}] + \mathbf{b}_{c_o}$.

Numerical guard. If $\text{Var}[\tilde{\mathbf{y}}_{c_o}]$ is near-zero (a near-dead channel), s_{c_o} explodes. We clip $s_{c_o} \in [0.5, 2.0]$ in the implementation.

6 Joint Bias & Variance Correction

When both corrections are applied, the variance scaling must pivot around the bias-corrected (FP32) mean rather than around the uncorrected quantized mean. Let

$$\mathbf{b}^{\text{bias}} = \mathbf{b} - \Delta\mathbf{b} \quad \text{and} \quad \mathbb{E}[\mathbf{y}_{c_o}] = \sum_{c_i} [\mathbf{W}_{\Sigma}]_{c_o, c_i} \mathbb{E}[\mathbf{x}_{c_i}] + \mathbf{b}_{c_o}, \quad (16)$$

where $\Delta\mathbf{b}$ is from (9) and $\mathbb{E}[\mathbf{y}_{c_o}]$ is the FP32 target mean. The variance scaling is then applied around $\mathbb{E}[\mathbf{y}_{c_o}]$:

$$\boxed{\widetilde{\mathbf{W}}_{c_o}^{\text{joint}} = s_{c_o} \widetilde{\mathbf{W}}_{c_o}, \quad \mathbf{b}_{c_o}^{\text{joint}} = s_{c_o}(\mathbf{b}_{c_o} - \Delta\mathbf{b}_{c_o}) + (1 - s_{c_o}) \mathbb{E}[\mathbf{y}_{c_o}].} \quad (17)$$

A direct check confirms that $\mathbb{E}[\mathbf{y}_{c_o}^{\text{joint}}] = \mathbb{E}[\mathbf{y}_{c_o}]$ and $\text{Var}[\mathbf{y}_{c_o}^{\text{joint}}] = \text{Var}[\mathbf{y}_{c_o}]$ under the working assumptions.

7 Computing $\mathbb{E}[\mathbf{x}]$ and $\text{Var}[\mathbf{x}]$ from Batch-Norm Parameters

When BN is applied before ReLU, the pre-ReLU output of channel c is approximately $\mathcal{N}(\mu = \beta_c, \sigma^2 = \gamma_c^2)$. The post-ReLU input to the next layer is then the *clipped normal* on $[a, b] = [0, \infty)$. With $\alpha = -\beta_c/\gamma_c$:

$$\mathbb{E}[\mathbf{x}_c] = \gamma_c \varphi(\alpha) + \beta_c(1 - \Phi(\alpha)), \quad (18)$$

$$\text{Var}[\mathbf{x}_c] = (\beta_c^2 + \gamma_c^2)(1 - \Phi(\alpha)) + \gamma_c \beta_c \varphi(\alpha) - \mathbb{E}[\mathbf{x}_c]^2, \quad (19)$$

where $\varphi(\cdot)$ is the standard normal PDF and $\Phi(\cdot)$ the standard normal CDF [1, App. C].

For the general clipped normal $f(X) = \text{clip}(X, a, b)$ with $X \sim \mathcal{N}(\mu, \sigma^2)$ and $\alpha = (a - \mu)/\sigma$, $\beta_n = (b - \mu)/\sigma$, $Z = \Phi(\beta_n) - \Phi(\alpha)$, the closed-form mean is

$$\mathbb{E}[f(X)] = \sigma(\varphi(\alpha) - \varphi(\beta_n)) + \mu Z + a\Phi(\alpha) + b(1 - \Phi(\beta_n)), \quad (20)$$

and the variance follows from the second-moment formula in [1, App. C]. The implementation handles the general case with $a = 0$, $b = \infty$ as the ReLU specialization.

8 Propagating Statistics Through Residual Connections and GAP

Within a ResNet block, $\mathbb{E}[\mathbf{x}]$ and $\text{Var}[\mathbf{x}]$ propagate as follows.

In-block conv inputs. For `conv2` (and `conv3` in a Bottleneck), the input is $\text{ReLU}(\text{BN}_{\text{prev conv}}(\cdot))$, so (18)–(19) apply with (β, γ) taken from the immediately preceding BN.

Residual sum. The output of the block is $\text{ReLU}(\text{BN}_{\text{main}}(\cdot) + \text{skip})$. Treating skip as approximately Gaussian and independent of the main branch (a simplification used by [1]), the pre-ReLU per-channel distribution is

$$\mathcal{N}(\beta_{\text{main}} + \mu_{\text{skip}}, \gamma_{\text{main}}^2 + \sigma_{\text{skip}}^2), \quad (21)$$

and the post-ReLU activation statistics follow from (18)–(19).

- *Identity skip:* $(\mu_{\text{skip}}, \sigma_{\text{skip}}^2)$ is the previous block’s post-residual stats (recursive).
- *Downsample skip:* the skip path is also BN_{ds} , contributing $(\beta_{\text{ds}}, \gamma_{\text{ds}}^2)$.

Global average pooling (fc input). The fully-connected layer of a ResNet sees $\bar{\mathbf{x}}_c = \frac{1}{HW} \sum_{i,j} \mathbf{x}_{c,i,j}$. Under the spatial-i.i.d. assumption,

$$\mathbb{E}[\bar{\mathbf{x}}_c] = \mathbb{E}[\mathbf{x}_c], \quad \text{Var}[\bar{\mathbf{x}}_c] = \text{Var}[\mathbf{x}_c] / (HW), \quad (22)$$

so the FC layer uses the layer-4 output mean unchanged and a variance scaled by $1/(HW)$ (e.g. $1/49$ for a 7×7 feature map at 224×224 input).

Stem. The first convolution receives the normalized image, not the output of a BN/ReLU. Analytical correction is undefined here and the stem layer is left uncorrected in the implementation.

9 Algorithm

Algorithm 1: Analytical bias / variance correction for a quantized model.

Input: FP32 model with BN; quantization method; correction mode

$\in \{\text{none}, \text{bias}, \text{variance}, \text{joint}\}.$

Output: Corrected quantized model.

$\{(\gamma_b, \beta_b)\}_b \leftarrow \text{capture_bn_params}(\text{model});$

fold BN into Conv (in place);

$\{(\mathbb{E}[\mathbf{x}], \text{Var}[\mathbf{x}])\}_L \leftarrow \text{compute_input_stats}$, propagating per §8;

for each weight layer L except stem do

 Save $\mathbf{W}_L^{\text{fp}};$

 Quantize: $\widetilde{\mathbf{W}}_L;$

$\epsilon_\Sigma \leftarrow (\widetilde{\mathbf{W}} - \mathbf{W}_L^{\text{fp}}).\text{sum}(\text{kernel axes});$

$\Delta \mathbf{b} \leftarrow \epsilon_\Sigma \cdot \mathbb{E}[\mathbf{x}]_L;$

$s \leftarrow \sqrt{(\mathbf{W}_\Sigma^{\text{fp}(2)} \text{Var}[\mathbf{x}]) / (\widetilde{\mathbf{W}}_\Sigma^{(2)} \text{Var}[\mathbf{x}])}$, clipped;

if $\text{mode} = \text{none}$ **then**

 | no change

else if $\text{mode} = \text{bias}$ **then**

 | $\mathbf{b} \leftarrow \mathbf{b} - \Delta \mathbf{b}$

else if $\text{mode} = \text{variance}$ **then**

 | $\mathbb{E}[\widetilde{\mathbf{y}}] \leftarrow \widetilde{\mathbf{W}}_\Sigma \cdot \mathbb{E}[\mathbf{x}] + \mathbf{b};$

 | $\widetilde{\mathbf{W}} \leftarrow s \odot \widetilde{\mathbf{W}};;$

 | $\mathbf{b} \leftarrow s \odot \mathbf{b} + (1 - s) \odot \mathbb{E}[\widetilde{\mathbf{y}}];$

else joint

 | $\mathbb{E}[\mathbf{y}] \leftarrow \mathbf{W}_\Sigma^{\text{fp}} \cdot \mathbb{E}[\mathbf{x}] + \mathbf{b};$

 | $\widetilde{\mathbf{W}} \leftarrow s \odot \widetilde{\mathbf{W}};;$

 | $\mathbf{b} \leftarrow s \odot (\mathbf{b} - \Delta \mathbf{b}) + (1 - s) \odot \mathbb{E}[\mathbf{y}];$

end

end

10 Implementation Notes

- All math is per-output-channel, so per-tensor and per-output-channel weight quantization are both supported without changes.
- Variance scaling is folded into the (already-quantized) weights and bias. After folding, the weights are no longer on the original symmetric quantization grid; this is acceptable for

an analysis pipeline that stores dequantized FP32 representations of quantized weights, but on real hardware s_{c_o} would more naturally be implemented as a per-output-channel rescale at the accumulator.

- The independence assumption between input channels is the same simplification used by [1] for $\mathbb{E}[\mathbf{x}]$. It is the dominant approximation; real activations exhibit modest cross-channel correlation, which would add cross terms to $\text{Var}[\mathbf{y}_{c_o}]$ that this derivation omits.
- The analytical clipped-normal model uses the BN running statistics. After fusion, the BN parameters $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ are captured beforehand; the running mean and variance are absorbed into the fused weight and bias.

References

- [1] Markus Nagel, Mart van Baalen, Tijmen Blankevoort, Max Welling. *Data-Free Quantization Through Weight Equalization and Bias Correction*. ICCV 2019. <https://arxiv.org/abs/1906.04721>